Compiled Problems – Regular Expressions

1. **2010-11**

   Which of the strings below are represented by the following regular expression?

   $$01^*0\ (01^*10^* \cup 01^*00^*1 \cup 10^*1)\ 11^*001^*0$$

   a) 0001101001 b) 0111001011 c) 010101010
   d) 0100000 e) 011001011 f) 01010010

2. **2011-12**

   Which of the following hexadecimal numbers, when written in binary, would be accepted by the following regular expression?

   $$010^*\ (10^*\ (01^* \cup 01^*0) \cup 11^*0)$$

   a) 48B b) 48E c) 5FE
   d) 67E e) 577 f) 60E

3. **2015-16**

   Given the following regular expression, simplify it using the identities to eliminate the union symbols.

   $$a\ (a^* \cup b)^*\ b\ (b^* \cup a)^*$$

4. **Mine**

   Which regular expressions are accepted by the regular expression: $a^*b\ (aab^* \cup baa^*)^*\ ba^*$?

   A) abbbbabab B) baabaaaaa C) aaaaaaabbaaaaaaaaaaa
   D) baabbbbaab E) bbbabbb F) abababaaba

5. **Mine**

   Which regular expressions are accepted by the following regular expression (u=A,…,Z and l=a,…,z)?

   ull*_u._ul*

   A) Ullyses_S._Grant B) Ms_ C._Blackstone C) Emmaus_H._School
   D) U_S._A E) Abe_Lincoln F) Madam_Im_Adam

6. **Mine**

Which regular expressions are equivalent to the following regular expression?

ab* (ba* ∪ ab*) ba*

A) ab*ba*ab*ba*
B) ab (b*a* ∪ b*ab) b*a*
C) ab*ba*ba* ∪ ab*ab*ba*
D) (ab*ba*ba*)*
E) ab* (ab* ∪ ba*) ba*
F) abb*a*ba* ∪ ab*abb*a*

7. **Mine**

Which regular expressions are accepted by the following regular expression (d=0,…,9)?

($ ∪ d*) d. (dd ∪ λ)

A) $0.45
B) $32.90
C) $8.456
D) 435.7
E) 5.43
F) $6.

8. **Mine**

Which regular expressions are equivalent to the following regular expression?

(ab*ba*)*

A) a*b*a*
B) (a*b*)*
C) ab*ba*
D) (ab*)*(ba*)*
E) (ab)*
F) a*b*a*

9. **Mine**

Which regular expressions are accepted by the following regular expression (f=a,…,m and s=n,…,z)?

ff*s*s ∪ (sf)*

A) fffffssssss
B) fsfsfsfs
C) sfsfsfsf
D) λ
E) sssssssffffff
F) fs

10. **Mine**

Which regular expressions are accepted by the following regular expression?

(1 ∪ 0)* 0* 0

A) all even numbers
B) all binary numbers
C) all starting with 1 and ending with 0
D) all odd numbers
E) all multiples of 4
F) all with alternating 0s and 1s